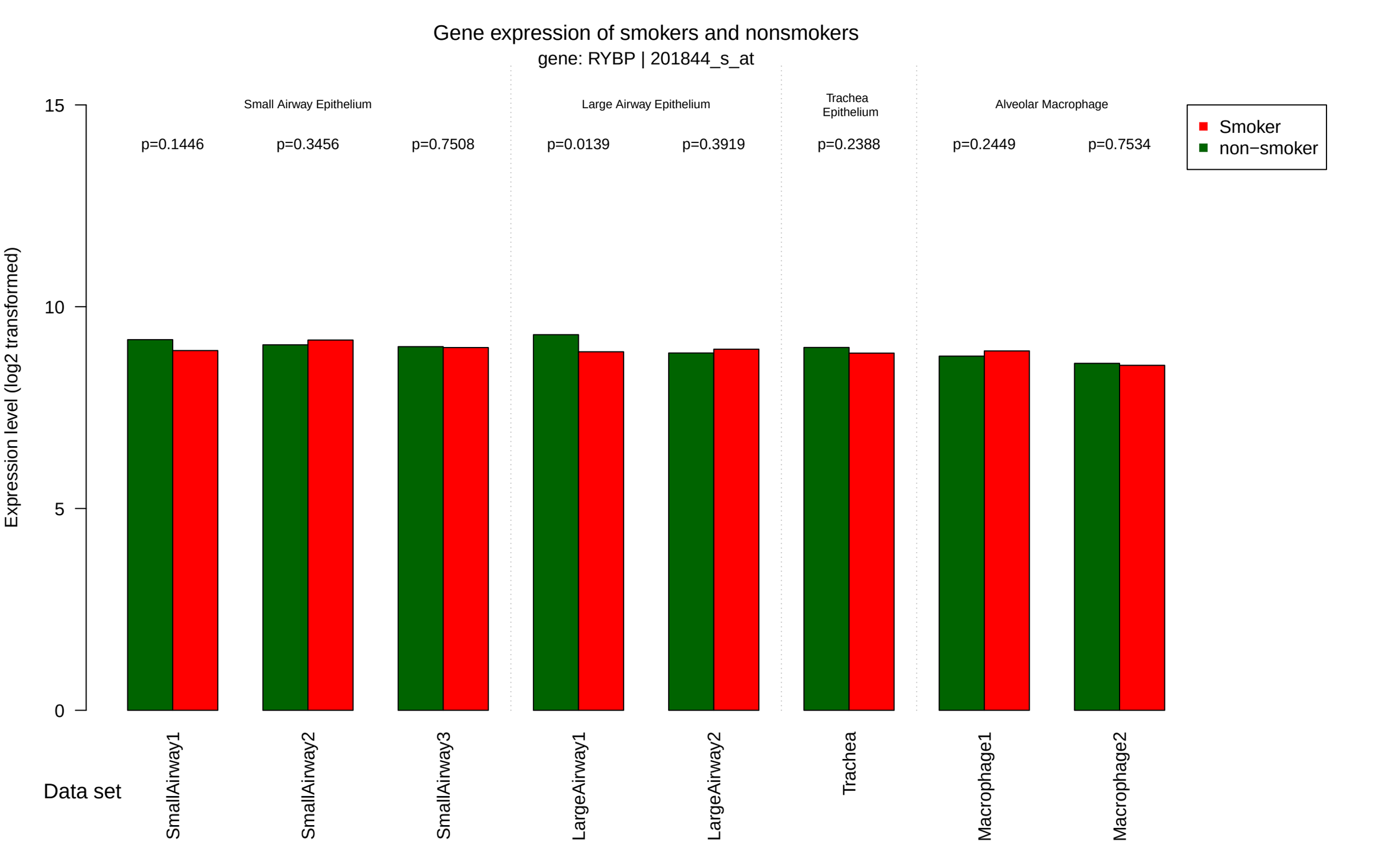
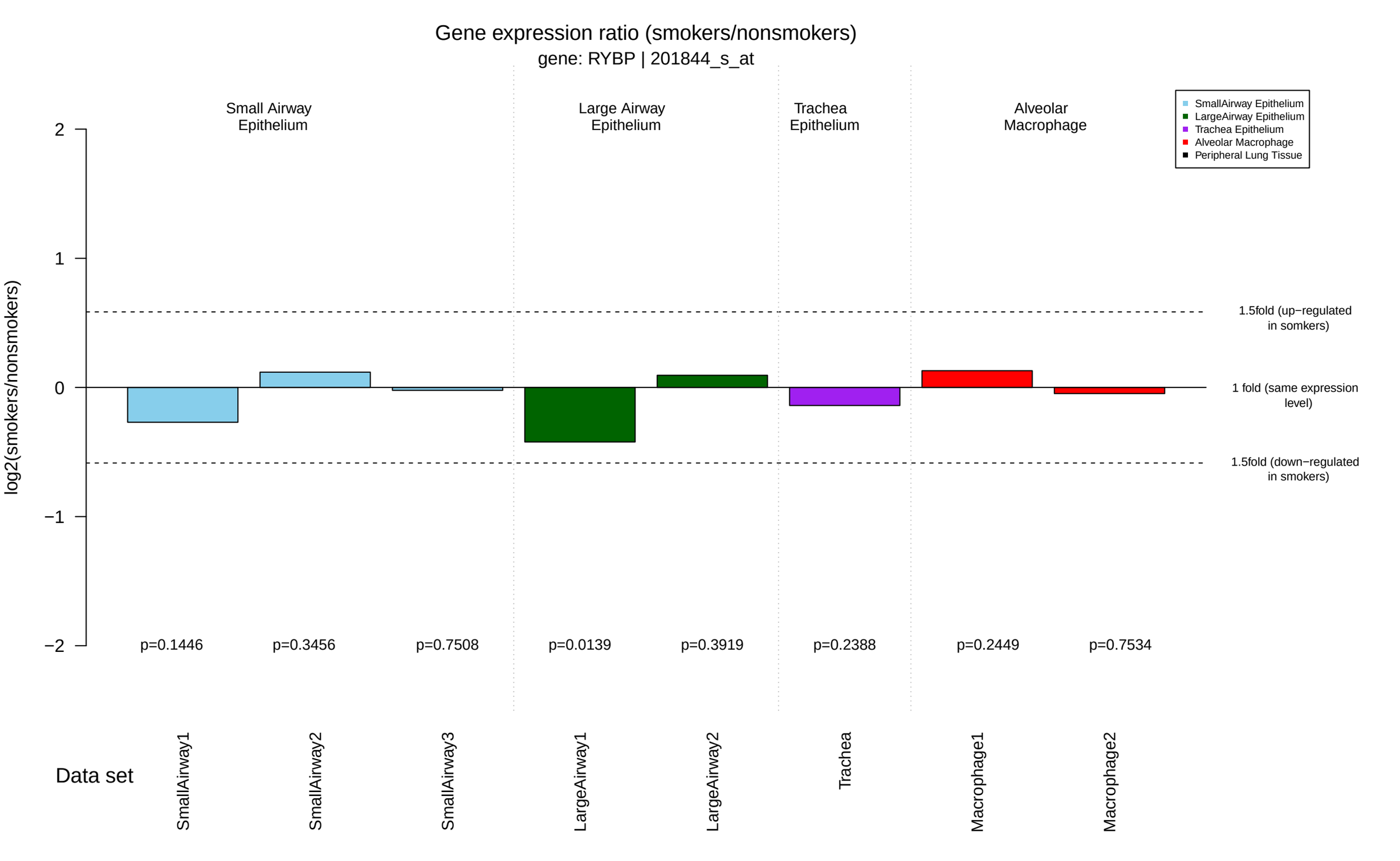
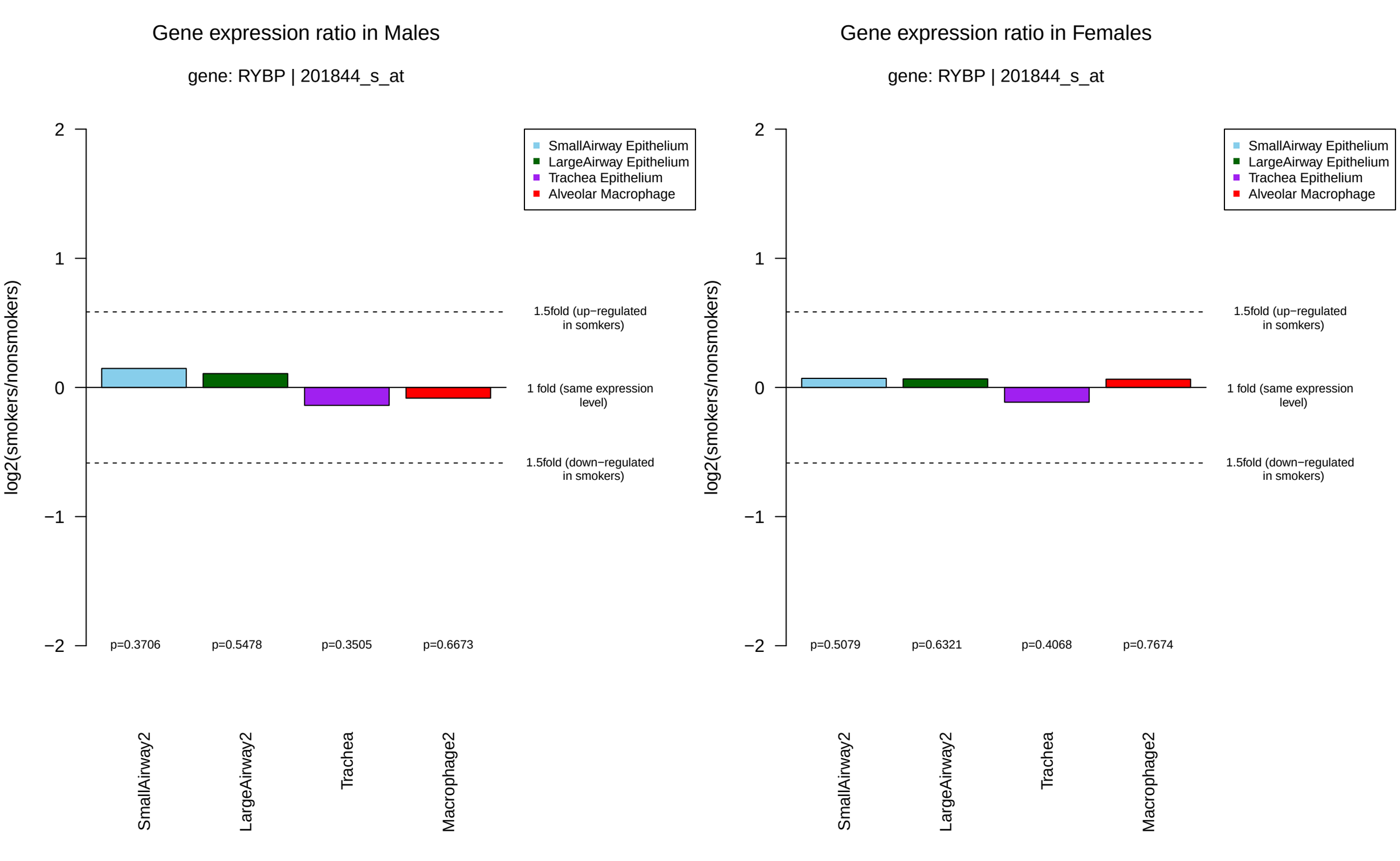
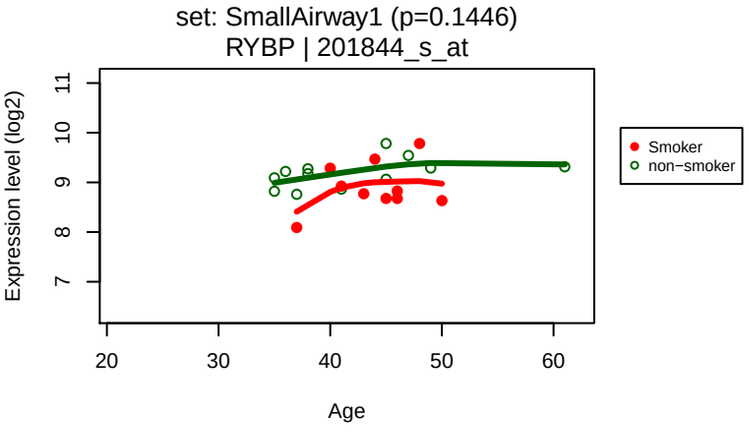
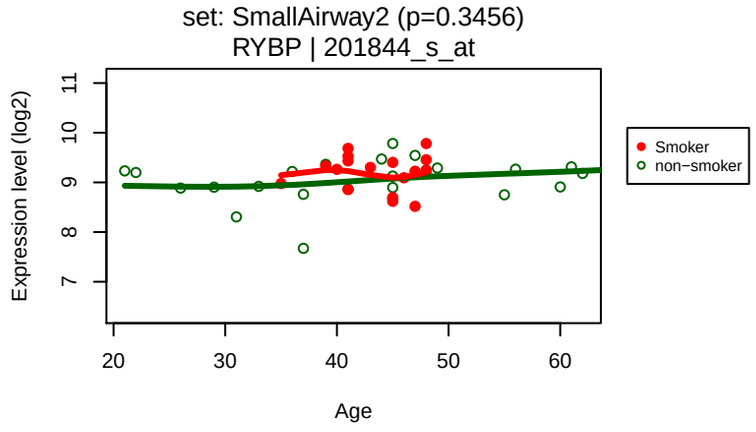
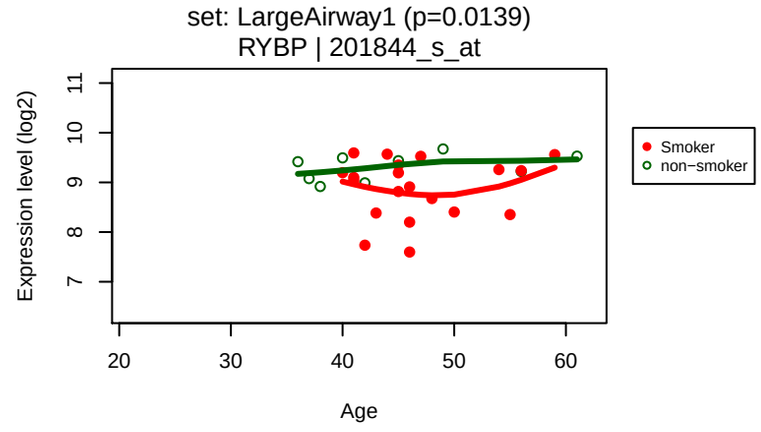

Gene expression of smokers and nonsmokers
gene: RYBP | 201844_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1446
p=0.3456
p=0.7508
p=0.0139
p=0.3919
p=0.2388
p=0.2449
p=0.7534
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

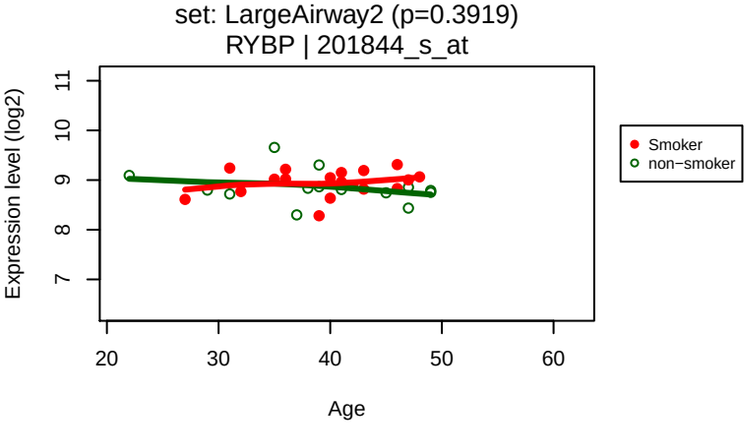

Gene expression ratio (smokers/nonsmokers)
gene: RYBP | 201844_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1446
p=0.3456
p=0.7508
p=0.0139
p=0.3919
p=0.2388
p=0.2449
p=0.7534
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

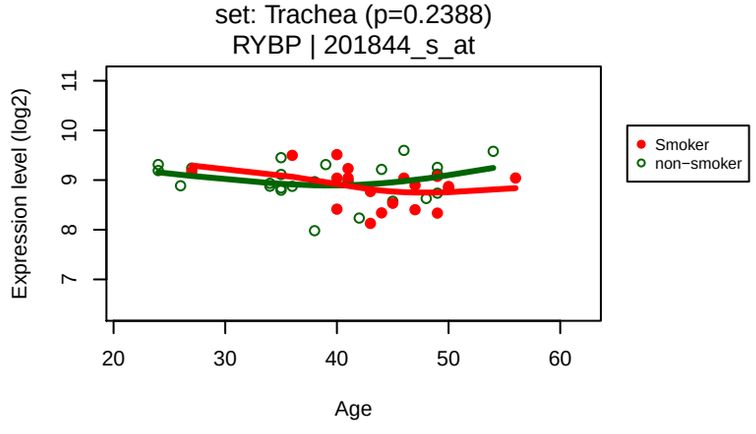

Gene expression ratio in Males
gene: RYBP | 201844_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3706
p=0.5478
p=0.3505
p=0.6673
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RYBP | 201844_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5079
p=0.6321
p=0.4068
p=0.7674
SmallAirway2
LargeAirway2
Trachea
Macrophage2

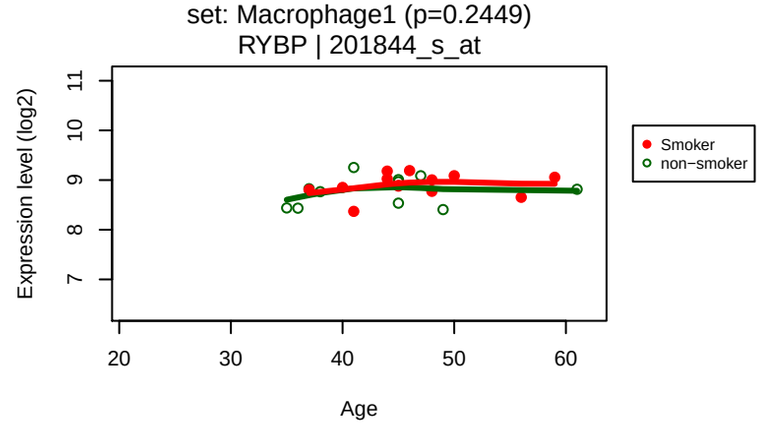

set: SmallAirway1 (p=0.1446)
RYBP | 201844_s_at
Expression level (log2)
Age
Smoker
non−smoker

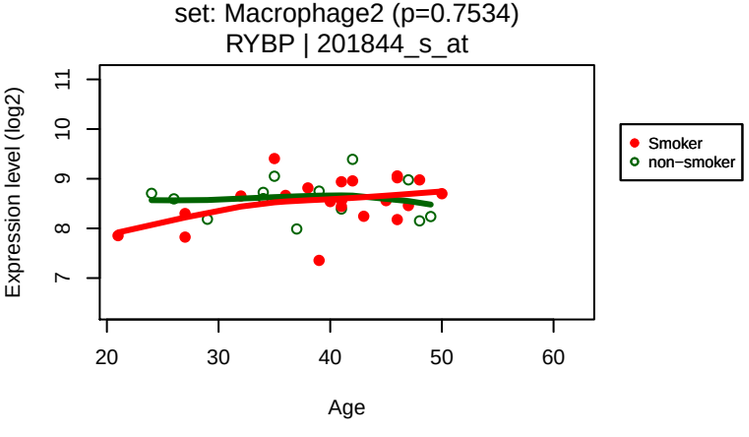

set: SmallAirway2 (p=0.3456)
RYBP | 201844_s_at
Expression level (log2)
Age
Smoker
non−smoker

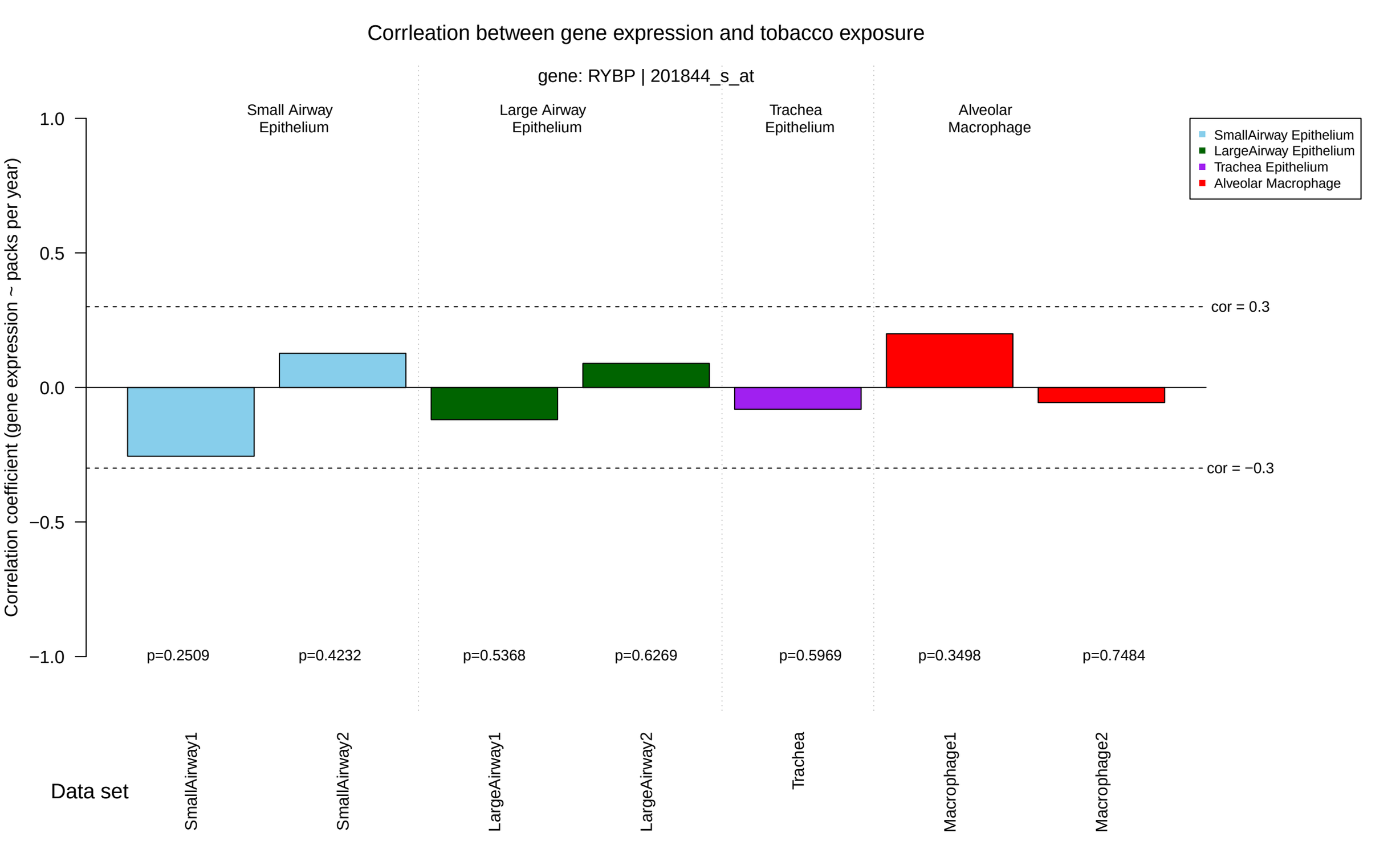

set: LargeAirway1 (p=0.0139)
RYBP | 201844_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3919)
RYBP | 201844_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2388)
RYBP | 201844_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2449)
RYBP | 201844_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7534)
RYBP | 201844_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RYBP | 201844_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2509
p=0.4232
p=0.5368
p=0.6269
p=0.5969
p=0.3498
p=0.7484
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2